11 Discrete Fourier Transform

Objectives

After covering this chapter, the user will be able to understand:

1. Relation between DFT and Fourier series.
2. Comparison between DFT and Fourier transform.
3. Properties of DFT.
4. Matrix formulation of DFT and IDFT.

Introduction

1. The DFT transform a DT sequence x[k] into a function $X(e^{j\omega})$ in the DTFT frequency domain ω. The independent variable ω is continuous and is confined to the range $-\pi \le \omega < \pi$.
2. Because of the continuous nature of ω, direct implementation of the DTFT is not suitable for such digital devices.
3. This chapter introduces the Discrete Fourier Transform (DFT), which can be computed efficiently on digital computers and other digital signal processing (DSP) boards.
4. The Discrete Fourier Transform (DFT) plays an important role in many applications of digital signal processing, including linear filtering, correlation analysis, and spectrum analysis.
5. A major reason for its importance is the existence of efficient algorithms for computing the DFT.

Discrete Fourier Transform (DFT):

Discrete Fourier Transform (DFT) is obtained by sampling one period of DTFT $X\left(e^{j\omega}\right)$ of x[n] at a finite number of frequency points. This sampling is performed for 'N' samplers over the range $0 \le \omega \le 2\pi, \omega_k = \frac{2\pi k}{N}$

Let x{n} be a finite-length sequence of length N, that is,

$x[n] = 0$ outside the range $0 \le n \le N-1$

The DFT of x[n], denoted as X[k], is defined by

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn} \qquad k = 0, 1, \ldots, N-1$$

Where W_N is the Nth root of unity given by

$$W_N = e^{-j(2\pi/N)}$$

The inverse DFT (IDFT) is given by

$$x[n] = \frac{1}{N} \sum_{n=0}^{N-1} X[k] W_N^{-kn} \qquad n = 0, 1, \ldots, N-1$$

The DFT pair is denoted by

$$x[n] \leftrightarrow X[k]$$

Important features of the DFT are the following:

1. There is a one-to-one correspondence between x[n] and X[k].
2. There is an extremely fast algorithm called the fast Fourier transform (FFT) for its calculation.
3. The DFT is closely related to the discrete Fourier series and the Fourier transform.
4. The DFT is the appropriate Fourier representation for digital computer realisation because it is discrete and of finite length in both the time and frequency domains.

Important points:

Zero padding:

To improve the resolution of the frequency axis ωin the DFT domain, a commonly used approach is to append the DT sequences with additional zero-valued samples. This process is called zero padding, and for an aperiodic sequence x[k] of length N is defined as follows:

$$x_{zp}[k] = \begin{cases} x[k] & ;0 \le k \le N-1 \\ 0 & ;N \le k \le M-1 \end{cases}$$

The zero-padded sequence $x_{zp}[k]$ has an increased length of *M*. The frequency

resolution $\Delta\omega$ of the zero-padded sequence is improved from 2π / N to 2π / M.

Relation between the DFT and the discrete Fourier series:

The discrete Fourier series representation of a periodic sequence x[n] with the fundamental period N_0 is given by

$$x[n] = \sum_{k=0}^{N_0-1} c_k e^{jk\Omega_0 n} \qquad \Omega_0 = \frac{2\pi}{N_0} \qquad \text{...(i)}$$

Where c_k are the Fourier coefficients

$$c_k = \frac{1}{N_0}\sum_{n=0}^{N_0-1} x[n]e^{-jk\Omega_0 n} \qquad \text{...(ii)}$$

The DFT of x[n], denoted as X[k], is defined by

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{kn} \qquad k = 0,1,\ldots,N-1 \qquad \text{...(iii)}$$

The inverse DFT (IDFT) is given by

$$x[n] = \frac{1}{N}\sum_{n=0}^{N-1} X[k]W_N^{-kn} \qquad n = 0,1,\ldots,N-1 \qquad \text{...(iv)}$$

Comparing eqs. (iv) and (iii) with eqs. (i) and (ii), we sees that X(k) of a finite sequence x{n} can be interpreted as the coefficients c_k in the discrete Fourier series representation of its periodic extension multiplied by the period N_0 and $N_0 = N$. That is

$$X[k] = Nc_k \qquad \text{...(v)}$$

Actually, the two can be made identical by including the factor 1/N with the DFT rather than with the IDFT.

Relationship between the DFT and the Fourier transform:

$$X(\Omega) = \mathcal{F}\{x[n]\} = \sum_{n=-\infty}^{\infty} x[n]e^{-j\Omega n} \qquad \text{...(vi)}$$

Let x[n] be a finite-length sequence of length N, that is,

$$x[n] = 0 \text{ outside the range } 0 \le n \le N-1 \qquad \text{...(vii)}$$

By definition (vi) the Fourier transform of x[n] defined by eq. (vii) can be expressed as

$$X(\Omega) = \sum_{n=0}^{N-1} x[n]e^{-j\Omega n} \qquad \text{...(viii)}$$

We see that

$$X[k] = X(\Omega)\Big|_{\Omega=k2\pi/N} = X\left(\frac{k2\pi}{N}\right) \qquad \text{...(ix)}$$

Thus, X[k] corresponds to the sampled $X(\Omega)$ at the uniformly spaced frequencies $\Omega = k2\pi / N$ for integer k.

1. Find 4 point DFT $x[n] = \{\underset{\uparrow}{0},1,2,3\}$

Solution:

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{\frac{-j2pk}{N}n}; k = 0 \text{ to } N-1$$

For 4 point DFT N=4

$$X[k] = \sum_{n=0}^{3} x[n]e^{\frac{-j2\pi k}{N}n}$$

$$X[k] = 0 + e^{-\frac{j\pi k}{2}} + 2e^{-j\pi k} + 3e^{-j\frac{3\pi}{2}k}$$

$$X[0] = 1+2+3 = 6$$

$$X[1] = e^{-\frac{j\pi}{2}} + 2e^{-j\pi} + 3e^{-j3\pi/2}$$

$$= -j - 2 + 3j = 2j - 2 = 2\sqrt{2} < -45º$$

$$X[2] = e^{-j\pi} + 2e^{-j2\pi} + 3e^{-j3\pi}$$

$$= -1 + 2 - 3 = -2 = 2\angle 180º$$

$$X[3] = e^{-j\frac{3\pi}{2}} + 2e^{-j3\pi} + 3e^{-j\frac{9\pi}{2}}$$

$$= j - 2 - 3j = -2 - 2j = 2\sqrt{2}\angle -135º$$

$$|X[k]| = \{\underset{\uparrow}{6}, 2\sqrt{2}, 2, 2\sqrt{2}\}$$

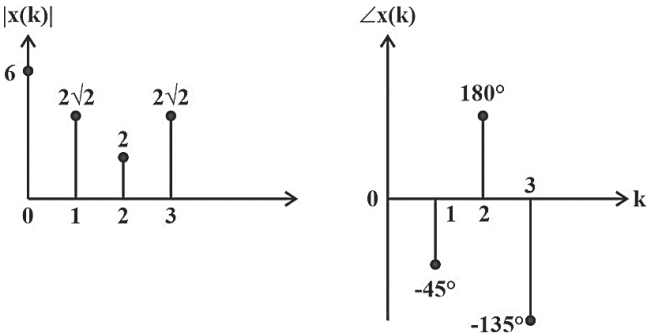


Fig. 11.1

Properties of DFT:

Linearity:

$$x_1[n] \xleftrightarrow{DFT} X_1[k] \;\&\; x_2[n] \xleftrightarrow{DFT} X_2[k]$$

Then $\alpha x_1[n] + \beta x_2[n] \xleftrightarrow{DFT} \alpha X_1[k] + \beta X_2[k]$

Proof:

$$DFT\left\{\alpha x_1[n]+\beta x_2[n]\right\}$$
$$=\sum_{n=0}^{N-1}\left(\alpha x_1[n]+\beta x_2[n]\right)e^{\frac{-j2\pi k}{N}n}$$
$$=\alpha\sum_{n=0}^{N-1}x_1[n]e^{\frac{-j2nk}{N}n}+\beta\sum_{n=0}^{N-1}x_2[n]e^{\frac{-j2\pi k}{N}n}$$

$$DFT\left\{\alpha x_1[n]+\beta x_2[n]\right\}=\alpha X_1[k]+\beta X_2[k]$$

Time shifting:

$$x\left[n\right]\xleftrightarrow{DFT}X\left[k\right]$$

Then $x\left[n-n_0\right]\xleftrightarrow{DFT}e^{-j\frac{2\pi}{N}kn_0}X\left[k\right]$

Proof:

$$DFT\left\{x\left[n-n_0\right]\right\}=\sum_{n=0}^{N-1}x\left[n-n_0\right]e^{-j\frac{2\pi}{N}/n}$$

$$Put\ \ n-n_0=m\Rightarrow n=m+n_0$$
$$DFT\ \ \left\{x\left[n-n_0\right]\right\}$$
$$=\sum_{m=-n_0}^{N-1-n_0}X[m]e^{-j\frac{2\pi}{N}k(m+n_0)}$$
$$=e^{-j\frac{2\pi}{N}kn_0}\sum_{m=-n_0}^{N-1-n_0}x[m]e^{-j\frac{2\pi}{N}km}$$
$$=e^{-\frac{2\pi}{N}kn_0}X[k]$$

Shift (in k domain):

If $x[n]\xleftrightarrow{DFT}X[k]$

Then $x\left[n\right]e^{j\frac{2\pi}{N}k_0n}\xleftrightarrow{DFT}X\left[k-k_0\right]$

Proof:

$$DFT\left\{x[n]e^{j\frac{2\pi}{N}k_0n}\right\}=\sum_{n=0}^{N-1}x[n]e^{-j\frac{2\pi}{N}\left[k-k_0\right]n}=X\left[k-k_0\right]$$

Convolution:

$$x_1[n]\xleftrightarrow{DFT}X_1[k]\ \&\ x_2[n]\xleftrightarrow{DFT}X_2[k]$$

$$Then\,x_1[n]*x_2[n]\xleftrightarrow{DFT}X_1[k]X_2[k]$$

Proof:

$$DFT\left\{x_1[n]*x_2[n]\right\}$$
$$=\sum_{n=0}^{N-1}\left[x_1[n]*x_2[n]\right]e^{-j\frac{2\pi}{N}kn}$$
$$=\sum_{n=0}^{N-1}\sum_{p=0}^{N-1}x_1[p]x_2[n-p]e^{-j\frac{2\pi}{N}kn}$$

$$DFT\left\{x_1[n]*x_2[n]\right\}=\sum_{p=0}^{N-1}x_1[p]\sum_{n=0}^{N-1}x_2[n-p]^{-j\frac{2\pi}{N}kn}$$

$$DFT\left\{x_1[n]*x_2[n]\right\}=\sum_{p=0}^{N-1}x_1[p]e^{-j\frac{2\pi}{N}pk}$$

$$\sum_{n=0}^{N-1}x_2[n-p]e^{-j\frac{2\pi}{N}(n-p)k}=X_1[k]X_2[k]$$

Conjugate:

If $x\left[n\right]\xleftrightarrow{DFT}X\left[k\right]$

Then $x^*[n]\xleftrightarrow{DFT}X^*[-k]$

Proof:

$$X[k]=\sum_{n=0}^{N-1}X[n]e^{-j\frac{2\pi}{N}kn}$$

Replace'k' with '-k'

$$X[-k]=\sum_{n=0}^{N-1}x[n]e^{j\frac{2\pi}{N}kn}$$

Conjugate both sides

$$X^*[-k]=\sum_{n=0}^{N-1}x^*[n]e^{-j\frac{2\pi}{N}kn}$$

$$x^*[n]\xleftrightarrow{DFT}X^*[-k]$$

Time reversal:

If $x\left[n\right]\xleftrightarrow{DFT}X\left[k\right]$

Then $x\left[-n\right]\xleftrightarrow{DFT}X\left[-k\right]$

Proof:

$$DFT\ \{x[-n]\}=\sum_{n=0}^{N-1}X[-n]e^{-\frac{2\pi}{N}kn}$$

Put – n = m

$$DFT\{x[-n]\}=\sum_{m=0}^{1-N}x[m]e^{\frac{2\pi}{N}km}$$
$$=\sum_{m=0}^{N-1}x[m]e^{-j\frac{2\pi}{N}(-k)m}=X[-k]$$

Parseval's relation:

If $x\left[n\right]\xleftrightarrow{DFT}X\left[k\right]$

Then $\sum_{n=0}^{N-1}\left|x[n]\right|^2=\frac{1}{N}\sum_{k=0}^{N-1}|x[k]|^2$

Proof:

$$\sum_{n=0}^{N-1}|x[n]|^2=\sum_{n=0}^{N-1}x\ [n]x^*[n]$$
$$=\sum_{n-0}^{N-1}\left[x[n]\frac{1}{N}\sum_{k-0}^{N-1}X^*[k]e^{-j\frac{2\pi}{N}kn}\right]$$

Adjusting the above equation

$$\sum_{n=0}^{N-1}|x[n]|^2=\frac{1}{N}\sum_{k=0}^{N-1}X^*[k]\sum_{n=0}^{N-1}x[n]e^{-j\frac{2\pi}{N}kn}$$

$$=\frac{1}{N}\sum_{k=0}^{N-1}X^*[k]X[k]=\frac{1}{N}|X[k]|^2$$

Linear convolution using DFT:

Let x[n] be a N_1 point sequence, and h[n] be N_2 point sequence. The linear convolution of x[n] and h[n] will produce a sequence y[n] of length N_1+N_2-1 samples

$$x[n]\to N_1;\quad h[n]\to N_2$$

$$\therefore y[n]\to\left(N_1+N_2-1\right)$$

Let X[k] be N_1+N_2-1 point DFT of x[n] and H[k] be N_1+N_2-1 point DFT of h[n] then

$$y\left[n\right]=IDFT\left\{X\left[k\right]H\left[k\right]\right\}$$

This technique of converting two finite duration sequences using DFT is known as 'Fast convolution', which is generally performed by using Fourier transform algorithms (FFT).

2. Find the linear convolution of $x\left[n\right]=\left\{1,0,5\right\}$ and $h\left[n\right]=\left\{0.5,1\right\}$

Solution:

$x\left[n\right]\to N_1=2$ and $h\left[n\right]\to N_2=2$

Then $y\left[n\right]\to N_1+N_2-1=3$ Samples

$X\left[k\right]\to N_1+N_2-1=3$ Samples

$$x\left[n\right]=\left\{1,0.5,0\right\}$$

$H\left[k\right]=3\to$ Samples

$$H\left[n\right]=\left\{0.5,1,0\right\}$$

$\therefore N=3$

$$X[k]=\sum_{n=0}^{2}x[n]e^{-j\frac{2\pi}{N}kn}$$

$$X[k]=1+0.5e^{-j\frac{2\pi}{3}k}$$

$X[0]=1.5$

$$X[1]=1+0.5e^{-j\frac{2\pi}{3}}=1+0.5\left\{-\frac{1}{2}-j\frac{\sqrt{3}}{2}\right\}$$

$$=0.75-j0.433$$

$$X[2]=1+0.5e^{-j\frac{4\pi}{3}}=0.75+j0.433$$

$$\therefore H(k)=\sum_{n=0}^{2}h[n]e^{-j\frac{2\pi}{N}kn}$$

$$H[k]=0.5+e^{-j\frac{2\pi}{3}kn}$$

$$H[0]=0.5+1=1.5$$

$$H[1]=0.5+e^{-j\frac{2\pi}{3}}=0.5+1\left\{-\frac{1}{2}-j\frac{\sqrt{3}}{2}\right\}=-j0.866$$

$$H[2]=0.5+e^{-j\frac{4\pi}{3}}=0.5+1\left\{-\frac{1}{2}+j\frac{\sqrt{3}}{2}\right\}=j0.866$$

$$\therefore Y[k]=X[k]H[k]$$

$$Y[0]=X[0]H[0]=1.5\times1.5=2.25$$

$$Y[1]=X[1]H[1]=-0.375-j0.6495$$

$$Y[2]=X[2]H[2]=-0.375+j0.6495$$

$$\therefore y[n]=\frac{1}{N}\sum_{k=0}^{N-1}Y[k]e^{j\frac{2\pi}{N}kn}=\frac{1}{3}\sum_{k=0}^{2}Y[k]e^{\frac{j-2\pi}{3}kn}$$

$$\therefore y[n]=\frac{1}{3}\left[Y[0]+Y[1]e^{j\frac{2\pi}{3}kn}+Y[2]e^{j\frac{4\pi}{3}kn}\right]$$

Put n = 0,1,2 we get

$$\therefore y\left[n\right]=\left\{0.5,1.25,0.5\right\}$$

Circular convolution:

To perform this convolution, at least one sequence should be periodic. If both the sequence are non- periodic, then periodically extend one of the sequences and perform circular convolution, i.e., sequences contain the same number of samples.

Representation of periodic signal:

4 point x(n) is given: $x\left(n\right)=\left\{1\quad2\quad3\quad4\right\}$

$x\left(n\right)=\left(x\left(n\right)\right)_N=x\left(n\right)$ modulo

$N=x\left(n+N\right)=x\left(n-N\right)$

1. $x(n)=\{1\ 2\ 3\ 4,1\ 2\ 3\ 4,1\ 2\ 3\ 4,\ 1\ 2\ 3\ 4\cdots\}$

2. $x(n) = \overbrace{1 \quad 2 \quad 3 \quad 4}$

3.

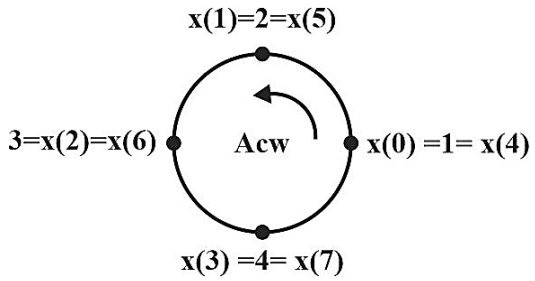


Fig. 11.2

$x(n)\{1,2,3,4\} \rightarrow g(n) = x(-n)$

$= \left(x(-n)_N\right) = x(N-n) = x(4-n)$

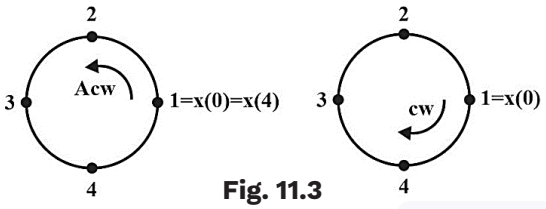


Fig. 11.3

$g(n) = \{1, \quad 4, \quad 3, \quad 2\}$

$g(0) \quad g(1) \quad g(2) \quad g(3)$

Note: While writing discrete-time signal from circle

1. Ref. position is observed → n = 0
2. Direction of an arrow is observed → $n = 1, n = 2, n = 3$, observed

$x(n) = \{\underset{\uparrow}{1}, 2, 3, 4\} \rightarrow x(n+1) = \{2, 3, 4, 1\}$

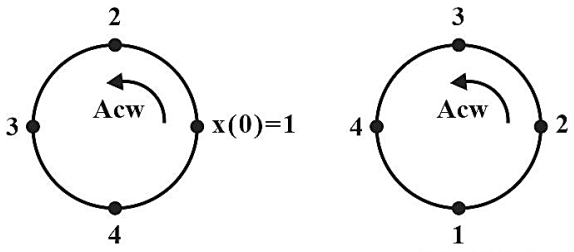


Fig. 11.4 g(n) {2,3,4,1}

Twiddle factor: or phase factor: $W_N = e^{-j\frac{2\pi}{N}}$

1. $W_N^0 = 1$
2. $W_N^N = 1$
3. $W_N^{\frac{N}{2}} = -1$
4. $W_N^{N+1} = W_N^1$
5. $W_N^{\frac{N}{2}+1} = -W_N^1$
6. $W_N^{n+N} = W_N^n$
7. $W_N^{N+\frac{N}{2}} = -W_N^n$
8. $W_N^{(n+\ell N)} = W_N^n \qquad W_N = e^{-\frac{j2\pi}{N}}$

   $\ell \in I$
9. $W_N^{\left((n+(2\ell+1)\frac{N}{2}\right)} = -W_N^n$

   $\ell \in I$
10. $W_N^{\ell N} = W_N^n = 1$

    $\ell \in I$
11. $W_N^{(2\ell+1)\frac{N}{2}} = -1$

    $\ell \in I$
12. $W_N = e^{-j2\pi/N} \Rightarrow W_N^* = e^{j2\pi/N}$

    $W_N^{-1} = W_N^*$

Matrix method of DFT/IDFT calculation:

1. DFT calculation

$$\underset{\text{Column matrix}}{[X(k)]} = \underset{\text{Square matrix}}{\left[W_N^K\right]} \quad \underset{\text{Column matrix}}{[x(n)]}$$

2. IDFT calculation

$$\underset{\text{Column matrix}}{[x[n]]} = \frac{\overset{\text{Square matrix}}{\left[W_N^k\right]^{-1}} \quad \overset{\text{Column matrix}}{[X(k)]}}{N}$$

Matrix method of DFT/IDFT calculation:

DFT: $[x(k)] = \left[W_N^n\right][x[n]]$

IDFT: $[x(n)] = \frac{1}{N}\left[W_N^n\right]^{-1}[x(k)] = \frac{1}{N}\left[W_N^n\right]^*[x(k)]$

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^* = \begin{bmatrix} A^* & B^* \\ C^* & D^* \end{bmatrix} = \begin{bmatrix} A^{-1} & B^{-1} \\ C^{-1} & D^{-1} \end{bmatrix}$$

A, B, C, D → Terms related to twiddle factor

Point DFT/IDFT: N = 2

Given

$x(n) = \{x(0), x(1)\} \quad 0 \leqslant n \leqslant 1$

$X(k) = \{X(0), X(1)\} \quad 0 \leqslant k \leqslant 1 \Rightarrow W_N$

$= W_2 = e^{\frac{-j2\pi}{2}} = -1$

DFT $[X(k)] = \left[W_N^n\right][x(n)]$ N = 2

$$\begin{bmatrix} X(0) \\ X(1) \end{bmatrix} = \begin{bmatrix} w_2^0 & w_2^0 \\ w_2^0 & w_2^1 \end{bmatrix}\begin{bmatrix} x(0) \\ x(1) \end{bmatrix} \Rightarrow \begin{bmatrix} X(0) \\ x(1) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} x(0) \\ x(1) \end{bmatrix}$$

$$[x(n)] = \frac{1}{N}\left[w_N^n\right]^{-1}[X(k)]$$

$$\begin{bmatrix} x(0) \\ x(1) \end{bmatrix} = \frac{1}{2}\begin{bmatrix} w_N^{-0} & w_N^{-0} \\ w_N^{-0} & w_N^{-1} \end{bmatrix}\begin{bmatrix} X(0) \\ X(1) \end{bmatrix}$$

$$\begin{bmatrix} x(0) \\ x(1) \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} X(0) \\ X(1) \end{bmatrix}$$

2. $x(n) = \{\underset{\uparrow}{1}, 1\}$ DFT of x(n)

Solution:

$$\begin{bmatrix} X(0) \\ X(1) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$X(0) = 2$

$X(1) = 0$

$X(k) = \{\underset{\uparrow}{2}, 0\}$

Point DFT/IDFT: N = 3

Given

$x(n) = \{x(0), x(1), x(2)\} \quad 0 \leqslant n \leqslant 2$

$X(k) = \{X(0), X(1), X(2)\} \quad 0 \leqslant k \leqslant 2$

$$w_N = w_3 = e^{\frac{-j2\pi}{3}} = \cos\frac{2\pi}{3} - j\sin\frac{2\pi}{3}$$

$$\begin{bmatrix} w_3^0 & w_3^0 & w_3^0 \\ w_3^0 & w_3^1 & w_3^2 \\ w_3^0 & w_3^2 & w_3^4 \end{bmatrix} \quad w_3^4 = \omega_3^3.\omega_3^1$$

DFT

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \end{bmatrix} = \begin{bmatrix} w_3^0 & w_3^0 & w_3 \\ w_3^0 & w_3^1 & w_3^2 \\ w_3^0 & w_3^2 & \underset{\downarrow}{w_3^4} \end{bmatrix}\begin{bmatrix} x(0) \\ x(1) \\ x(2) \end{bmatrix}$$

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & w_3^1 & w_3^2 \\ 1 & w_3^2 & w_3^1 \end{bmatrix}\begin{bmatrix} x(0) \\ x(1) \\ x(2) \end{bmatrix}$$

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & w_3^{-1} & w_3^{-2} \\ 1 & w_3^{-2} & w_3^{-4} \end{bmatrix}\begin{bmatrix} X(0) \\ X(1) \\ X(2) \end{bmatrix}$$

4 point DFT/IDFT: N = 4

Given $x(n) = \{x(0), x(1), x(2), x(3)\}$

$X(k) = \{X(0), X(1), X(2), X(3)\}$

$$W_N = W_4 = e^{-\frac{j2\pi}{4}} = -j$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & w_4^1 & w_4^2 & w_4^3 \\ 1 & w_4^2 & w_4^4 & w_4^6 \\ 1 & w_4^3 & w_4^6 & w_4^9 \end{bmatrix} \overset{DFT}{=} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & +j \\ 1 & -1 & 1 & -1 \\ 1 & +j & -1 & -j \end{bmatrix}$$

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & w_4^1 & w_4^2 & w_4^3 \\ 1 & w_4^2 & w_4^4 & w_4^6 \\ 1 & w_4^3 & w_4^6 & w_4^9 \end{bmatrix}\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

DFT

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & +j \\ 1 & -1 & 1 & -1 \\ 1 & +j & -1 & -j \end{bmatrix}\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix}$$

IDFT

$$\begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \end{bmatrix} = \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & +j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix}\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix}$$

$$\boxed{\begin{matrix} X(0) = 2 \\ X(1) = 2 + 2j \\ X(2) = -2 \\ X(3) = 2 - 2j \end{matrix}} \quad X(k) = \{2, 2 + 2j, -2, 2 - 2i\}$$

3. Calculate 4 point DFT of $x(n) = \{1, 0, -1, 2\}$

Solution:

$$\begin{bmatrix} X(0) \\ X(1) \\ X(2) \\ X(3) \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & +j \\ 1 & -1 & 1 & -1 \\ 1 & +j & -1 & -j \end{bmatrix}\begin{bmatrix} 1 \\ 0 \\ -1 \\ 2 \end{bmatrix}$$

Note:

1. $X(k) = \sum_{n=0}^{N-1} \frac{x(n)}{P} W_N^{-kn}$

   If $x(-n) = x(n)$

   DFT $\left[DFT\left(x(n)\right)\right] = \left(\sqrt{N}\right)^2 \frac{\{x(n)\}}{p^2}$

2. $x[n] = \frac{q}{N}\sum_{k=0}^{N-1} X(k)W_N^{+kn} = \sum_{k=0}^{N-1}\frac{qX(k)}{N}W_N^{kn}$

$$IDFT\left[IDFT\left\{X\left(k\right)\right\}\right] = \frac{1}{\left(\sqrt{N}\right)^2}\left\{X\left(k\right)\right\}$$

$$= \frac{1}{\left(\frac{\sqrt{N}}{q}\right)^2}\left\{X\left(k\right)\right\}$$

If $X\left(-k\right) = X\left(k\right)$ $IDFT\left\{IDFT\left\{X\left(k\right)\right\}\right\}$

$$= q^2\left\{\frac{X\left(l\right)}{\left(\sqrt{N}\right)^2}\right\}$$

Case a) If $x\left(-n\right) = x\left(n\right)$ $X(k) = \sum_{n=0}^{N-1}\frac{x(n)}{P}e^{\frac{-j2\pi}{N}kn}$

$$DFT\left[DFT(x(n)\right] = \frac{\left(\sqrt{N}\right)^2}{p^2}\{x(n)\}$$

$$DFT\left[DFT[DFT(x(n)\right] = DFT\left[\frac{\left(\sqrt{N}\right)^2}{P^2}\{x(n)\}\right]$$

$$DFT\left[DFT[DFT[DFT(x(n)]]\right] = \frac{\left(\sqrt{N}\right)^4}{p^4}\{x(n)\}$$

Case b) $x(n) = \frac{Q}{N}\sum_{k=0}^{N-1} X(k)e^{\frac{j2\pi}{N}kn}$

$X\left(-k\right) = X\left(k\right)$

$$IDFT\left[IDFT\{X(k)\}\right] = \frac{Q^2}{\left(\sqrt{N}\right)^2}\{X(k)\}$$

$$IDFT\left[IDFT\ [IDFT\{X(k)\}]\right] = IDFT\frac{\left[Q^2\left\{X(k)\right\}\right]}{\left(\sqrt{N}\right)^2}$$

$$IDFT\ [IDFT\ [IDFT\ [IDFT\{X(k)\}]]] = \frac{Q^4}{\left(\sqrt{N}\right)^4}\left\{X(k)\right\}$$

Test Your Understanding

1. x(n) is 4 point DFT where

$$x(n) = \frac{1}{\sqrt{N}}\sum_{k=0}^{N-1} x[k]e^{\frac{j2\pi kn}{N}}$$

$IDFT\ [IDFT\ [IDFT\ [IDFT\{X(k)\}]]] = ?$

Solution:

$\{X(k)\}$

11.2. Perform the circular convolution $x_1\left[n\right] = \left\{2,1,2,1\right\}$ and $x_2\left[n\right] = \left\{1,2,3,4\right\}$.

Solution:

First method:

$x_3[m] = x_1[n] * x_2[n]$

$$x_3[m] = \sum_{n=0}^{N-1} x_1[n]x_2[n-m]$$

$\therefore m = 0$ to $N-1 \Rightarrow m = 0 to 3$ $[\because N = 4]$

For $m = 0$

$$x_3[0] = \sum_{n=0}^{3} x_1[n]\left[x_2[-n]\right]$$

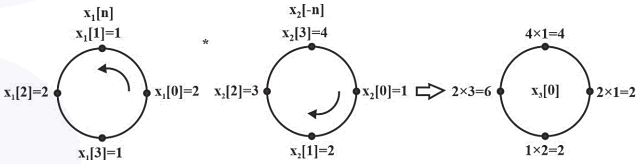


Fig. 11.5

$x_3[0] = 2 + 4 + 6 + 2 = 14$

For m = 1

$$x_3[1] = \sum_{n=0}^{3} x_1[n]\left[x_2[1-n]\right]$$

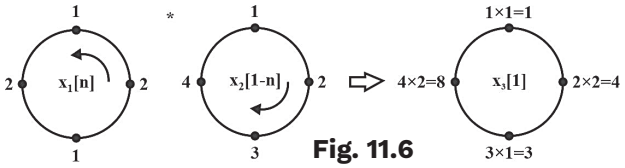


Fig. 11.6

$x_3[1] = 4 + 1 + 8 + 3 = 16$

For $m = 2$

$$x_3[2] = \sum_{n=0}^{3} x_1[n]\left[x_2[2-n]\right]$$

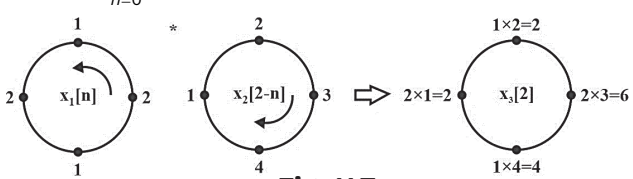


Fig. 11.7

$\therefore x_3\left[2\right] = 6 + 2 + 2 + 4 = 14$

For $m = 3$

$$x_3\left[3\right] = \sum_{n=0}^{3} x_1\left[n\right]\left[x_2[3-n]\right]$$

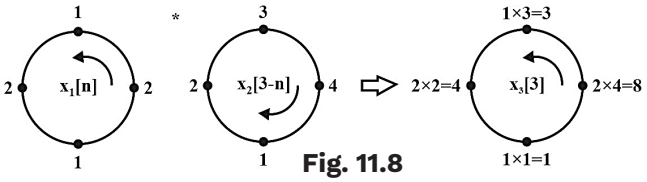


Fig. 11.8

$\therefore x_3[3] = 8 + 3 + 4 + 1 = 16$

$\therefore x_1[n] * x_2[n] = \{14, 16, 14, 16\}$

Second method:

$x_1[n] = \{2, 1, 2, 1\}$

$$X_1[k] = 2e^{-j\frac{2\pi}{4}k\times 0} + e^{-j\frac{2\pi}{4}k\times 1} + 2e^{-j\frac{2\pi}{4}k\times 2} + e^{-j\frac{2\pi}{4}k\times 3}$$

$X_1[k] = \{6, 0, 2, 0\}$

$x_2[n] = \{1, 2, 3, 4\}$

$$X_2[k] = e^{-j\frac{2\pi}{4}k\times 0} + 2e^{-j\frac{2\pi}{4}k\times 1} + 3e^{-j\frac{j\pi}{4}k\times 2} + 4e^{-j\frac{2\pi}{4}k\times 3}$$

$X_2[k] = \left\{10, -2 + 2j,\ -2, -2 - 2j\right\}$

$x_3[n] = x_1[n] * x_2[n]$

$X_3[k] = X_1[k]X_2[k]$

$X_3[0] = X_1[0]X_2[0] = 60$

$X_3[1] = X_1[1]X_2[1] = 0$

$X_3[2] = X_1[2]X_2[2] = -4$

$X_3[3] = X_1[3]X_2[3] = 0$

$$x_3[n] = \frac{1}{N}\sum_{k=0}^{N-1} X_3[k]e^{j\frac{2\pi}{N}kn}$$

$$x_3[n] = \frac{1}{4}\sum_{k=0}^{3} X_3[k]e^{j\frac{2\pi}{4}kn} = \frac{1}{4}\left[60 - 4e^{j\frac{2\pi}{4}(2)n}\right]$$

$\therefore x_3[0] = 14, x_3[1] = 16, x_3[2] = 14, x_3[3] = 16$

$x_3[n] = \{14, 16, 14, 16\}$

Matrix method:

$$\begin{bmatrix} x_2[0] & x_2[3] & x_2[2] & x_1[1] \\ x_2[1] & x_2[0] & x_2[3] & x_2[2] \\ x_2[2] & x_2[1] & x_2[0] & x_2[3] \\ x_2[3] & x_2[2] & x_2[1] & x_2[0] \end{bmatrix}\begin{bmatrix} x_1[0] \\ x_1[1] \\ x_1[2] \\ x_1[3] \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 & 3 & 2 \\ 2 & 1 & 4 & 3 \\ 3 & 2 & 1 & 4 \\ 4 & 3 & 2 & 1 \end{bmatrix}_{4\times 4}\begin{bmatrix} 2 \\ 1 \\ 2 \\ 1 \end{bmatrix}_{4\times 1} = \begin{bmatrix} 2+4+6+2 \\ 4+1+8+3 \\ 6+2+2+4 \\ 8+3+4+1 \end{bmatrix} = \begin{bmatrix} 14 \\ 16 \\ 14 \\ 16 \end{bmatrix}$$

$x_1[n] * x_2[n] = \{14, 16, 14, 16\}$

Table method:

$x[n] = \{2, 1, 2, 1\}, h[n] = \{1, 2, 3, 4\}$

n				0	1	2	3		
x[n]				2	1	2	1		
h[n]				1	2	3	4		
h[–n]	4	3	2	1	4	3	2	1	
h[1–n]		4	3	2	1	4	3	2	1
h[2–n]			4	3	2	1	4	3	2
h[3–n]				4	3	2	1	4	3

Table 11.1

$$y[m] = x[n] * h[n] = \sum_{n=0}^{N-1} x[n][h[m-n]]$$

$$y[0] = \sum_{n=0}^{3} x[n][h[-n]] = 2 + 4 + 6 + 2 = 14$$

$$y[1] = \sum_{n=0}^{3} x[n][h[1-n]] = 4 + 1 + 8 + 3 = 16$$

$$y[2] = \sum_{n=0}^{3} x[n][h[2-n]] = 6 + 2 + 2 + 4 = 14$$

$$y[3] = \sum_{n=0}^{3} x[n][h[3-n]] = 8 + 3 + 4 + 1 = 16$$

Common mistake:

1. Most common mistake is of +ve and -ve sign while writing DFT matrix.
2. Another mistake is related to shifting during circular convolution.

Test Your Understanding

1. Let x[n] be a finite length sequence given by

$x[n] = \{2, 1, 1, 0, 3, 2, 0, 3, 4, 6\}$, whose DFT is X(k). Evaluate the following without actually calculating X[k].

(A) X[0]

(B) X[5]

(C) $\sum_{k=0}^{9} X[k]$

(D) $\sum_{k=0}^{9} |X[k]|^2$

Solution:

(A) X[0] = 22

(B) X[5] = -2

(C) $\sum_{k=0}^{9} X[k] = 20$

(D) $\sum_{k=0}^{9} |X[k]|^2 = 800$

Exercise Level 1

1. The number of complex additions and multiplication in FFT are, respectively,

(A) $\frac{N}{2}\log_2 N$ *and* $N\log_2 N$

(B) $N\log_2 N$ *and* $\frac{N}{2}\log_2 N$

(C) $\frac{N}{2}\log_2 N$ *and* $\log_2 N$

(D) $\log_2 N$ *and* $\frac{N}{2}\log_2 N$ **[ESE-2017]**

2. If the complex multiply operation takes 1 μs, the time taken to compute 1024-point DFT directly will be nearly **[ESE-2019]**

(A) 3.45 s

(B) 2.30 s

(C) 1.05 s

(D) 0.60 s **[Set-02]**

Exercise Level 2

1. The 4-point Discrete Fourier Transform (DFT) of a discrete-time sequence [1, 0, 2, 3] is

(A) [0, -2 + 2j, 2, -2, -2 -2j]

(B) [2, 2 + 2j, 6, 2 - 2j]

(C) [6, 1 - 3j, 2, 1 + 3j]

(D) [6, -1 + 3j, 0, -1-3j]

2. x(n) is a real-valued periodic sequence with a period N. x(n) and X(k) form N-point Discrete Fourier Transform (DFT) pairs. The DFT Y(k) of the sequence $y(n) = \frac{1}{N}\sum_{r=0}^{n-1} x(r)\,x(n+r)$ is

(A) $|X(k)|^2$

(B) $\frac{1}{N}\sum_{r=0}^{N-1} X(r)X^*(k+r)$

(C) $\frac{1}{N}\sum_{r=0}^{N-1} X(r)X(k+r)$

(D) 0

3. The DFT of a vector $\begin{bmatrix} a & b & c & d \end{bmatrix}$ is the vector $\begin{bmatrix} \alpha & \beta & \gamma & \delta \end{bmatrix}$. Consider the product

$$\begin{bmatrix} p & q & r & s \end{bmatrix} = \begin{bmatrix} a & b & c & d \end{bmatrix} \begin{bmatrix} a & b & c & d \\ d & a & b & c \\ c & d & a & b \\ b & c & d & a \end{bmatrix}$$

The DFT of vector $\begin{bmatrix} p & q & r & s \end{bmatrix}$ is a scaled version of

(A) $\begin{bmatrix} \alpha^2 & \beta^2 & \gamma^2 & \delta^2 \end{bmatrix}$

(B) $\begin{bmatrix} \sqrt{\alpha} & \sqrt{\beta} & \sqrt{\gamma} & \sqrt{\delta} \end{bmatrix}$

(C) $\begin{bmatrix} \alpha+\beta & \beta+\delta & \delta+\gamma & \gamma+\alpha \end{bmatrix}$

(D) $\begin{bmatrix} \alpha & \beta & \gamma & \delta \end{bmatrix}$

Answer Keys

Level 1					
1	(A)	2	(C)		
Level 2					
1	(D)	2	(A)	3	(A)

Table 11.2

Solutions Level – 1

1. (B)

For radix 2FFT, there are $\log_2 N$ stages, and each state has $N/2$ complex multiplication and N complex addition. Thus the total number of complex multiplication will be $\frac{N}{2}\log_2 N$ and the number of complex addition will be $N\log_2 N$.

2. (C)

$N = 1024$

Time for one multiplication $= 1\mu\,\text{sec}$

Total time DFT $= N^2 \times 1\mu\,\text{sec}$

$= 1024^2 \times 10^{-6}\,\text{sec}$

$= 1.05\,\text{sec}$

Solutions Level – 2

1. (D)

Given: $x[n] = \{1, 0, 2, 3\}$

4-Point DFT of the signal $x[n]$ is given by.

$X(k) = Wx[n]$

Where, $W = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix}$

Since, $W_N = e^{\frac{-j2\pi}{N}}$

Where, $W_4 = e^{\frac{-j2\pi}{4}} = e^{\frac{-j\pi}{2}}$

$W_4 = \cos\left(\frac{\pi}{2}\right) - j\sin\left(\frac{\pi}{2}\right)$

$W_4 = -j$

$W = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$

Hence, $X(k) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \\ 3 \end{bmatrix}$

$X(k) = \begin{bmatrix} 1 & +0 & +2 & +3 \\ 1 & -0 & -2 & +j3 \\ 1 & -0 & +2 & -3 \\ 1 & +0 & -2 & -j3 \end{bmatrix}$

$X(k) = \begin{bmatrix} 6 \\ -1 + j3 \\ 0 \\ -1 - j3 \end{bmatrix}$

Hence, the correct option is (D).

2. (A)

Given: $y[n] = \frac{1}{N}\sum_{r=0}^{N-1} x[r]x[n+r]$

Above equation represents the autocorrelation of signal x[n] in the discrete-time domain.

Discrete Fourier transform of the auto correlation function is given by,

$F\{y[n]\} = X(k)X^*(k)$

$F\{y[n]\} = |X(k)|^2$

$F\{y[n]\} = \text{Power spectral density of } x[n]$

Hence, the correct option is (A).

3. (A)

Given:

DFT of the vector $[a \quad b \quad c \quad d]$ is $[\alpha \quad \beta \quad \gamma \quad \delta]$,

i.e $$\begin{bmatrix}\alpha\\ \beta\\ \gamma\\ \delta\end{bmatrix} = \begin{bmatrix}1 & 1 & 1 & 1\\ 1 & -j & -1 & j\\ 1 & -1 & 1 & -1\\ 1 & j & -1 & -j\end{bmatrix}\begin{bmatrix}a\\ b\\ c\\ d\end{bmatrix}$$

$$\begin{bmatrix}a\\ b\\ \gamma\\ \delta\end{bmatrix} = \begin{bmatrix}a+b+c+d\\ a-jb-c+jd\\ a-b+c-d\\ a+jb-c-jd\end{bmatrix} \quad \text{...(i)}$$

Now, it is given that,

$$[p \quad q \quad r \quad s] = [a \quad b \quad c \quad d]\begin{bmatrix}a & b & c & d\\ d & a & b & c\\ c & d & a & b\\ b & c & d & a\end{bmatrix}$$

$$= [a^2 + bd + c^2 + bd \quad ab + ab + cd + cd \quad ac + b^2 + ac + +d^2 \quad ad + bc + bc + ad]$$

$$= [a^2 + c^2 + 2bd \quad 2bd + 2cd \quad b^2 + d^2 + 2ac \quad 2ad + 2bc] \quad \text{...(ii)}$$

DFT of $[p \quad q \quad r \quad s]$ is given as,

$$= [p \quad q \quad r \quad s]\begin{bmatrix}1 & 1 & 1 & 1\\ 1 & -j & -1 & j\\ 1 & -1 & 1 & -1\\ 1 & j & -1 & -j\end{bmatrix}$$

$$= [p+q+r+s \quad p-jq-r+js \quad p-q+r-s \quad p+jq-r-js]$$

From equation (ii)

$$p+q+r+s = (a^2 + c^2 + 2bd) + (2ab + 2cd) + (b^2 + d^2 + 2ac) + (2ad + 2bc)$$

$$p+q+r+s = (a+b+c+d)^2$$

$$p+q+r+s = \alpha^2 \quad \text{[From eqution (i)]}$$

Clearly, DFT of $[p \quad q \quad r \quad s]$ is proportional to $[a^2 \quad \beta^2 \quad \gamma^2 \quad \delta^2]$.

Hence, the correct option is (A).